

802.11be vs 802.11n: P-EDCA Tail-Latency Comparison

CWds × QSRC × PSRC parameter sweep | CBR / Poisson / MMPP / OnOff traffic | P-EDCA STA perspective

Simulation setup

Topology	30 STAs → 1 AP, random 1-5 m disc; 5 GHz ch36 / 20 MHz
Traffic	Uplink AC_VO UDP, 1 Mbps per STA (1000 B); CBR / Poisson / MMPP / OnOff
PHY	11n = HtMcs7 (65 Mbps) vs 11be = EhtMcs7 (73.1 Mbps @ 3.2 μs GI)
Control	OFDM 6 Mbps for both; A-MPDU on; VO TXOP limit 2.08 ms
Fairness	The two simulation programs differ by exactly 2 lines (SetStandard / DataMode)
Sweep	nPedca ∈ {5, 15, 30}; CWds{0,1} × QSRC{0-5} × PSRC{1-3} = 36 combos
Statistics	10 seeds × 10 s per point; plus an EDCA-only baseline (nPedca = 0)

Data integrity: 4 traffic × 290 delay histograms are independent runs (11n / 11be md5 all differ)

One-line takeaway

11be's edge is not faster per-frame airtime (nearly identical) - it pulls the system back from the high-collision operating point: tail latency more than halves under load, and the P-EDCA mechanism becomes worthwhile again.

P99 -56% to -78%

Tail latency at full P-EDCA load (30/30), best params
Consistent across all 4 traffic models; 4.5-9.5 ms vs 17-21 ms

Loss 7-9% → 1-4%

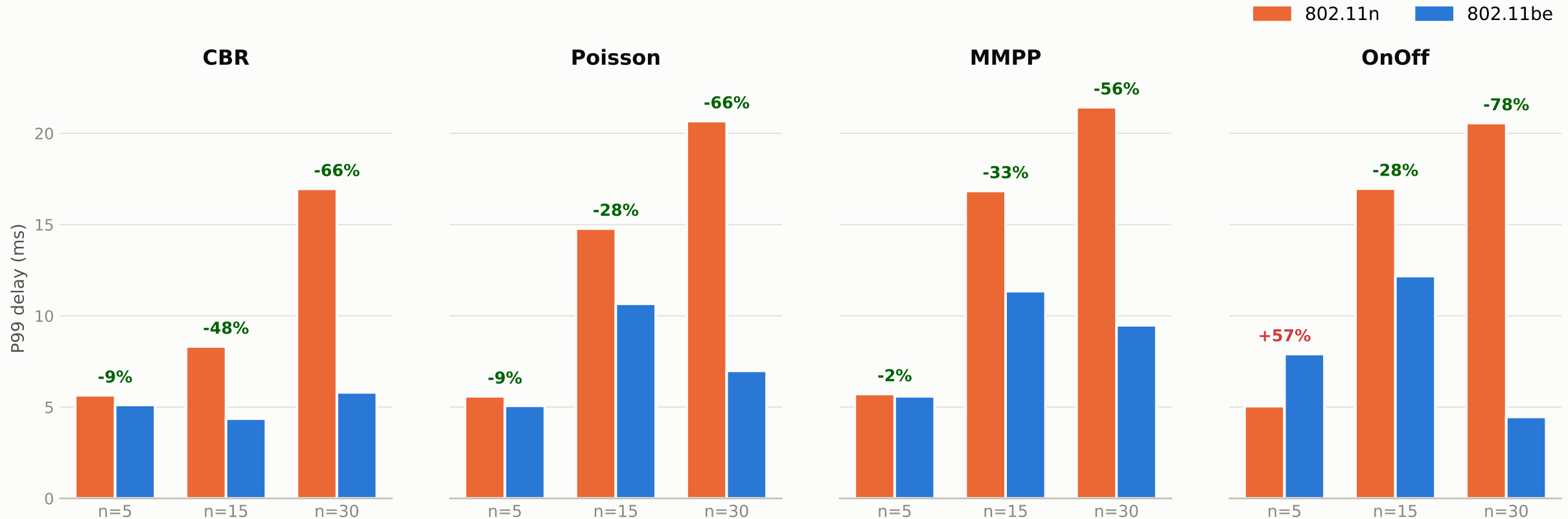
VO retry-limit loss, EDCA-only baseline
Same parameters and load; 11n stuck in collision feedback, 11be not

+44% to +75% vs EDCA

P-EDCA still pays off at full penetration on 11be
Only +3% to +13% on 11n - P-EDCA value grows with the PHY generation

Core result: P-EDCA P99 latency — 11be's edge grows with load

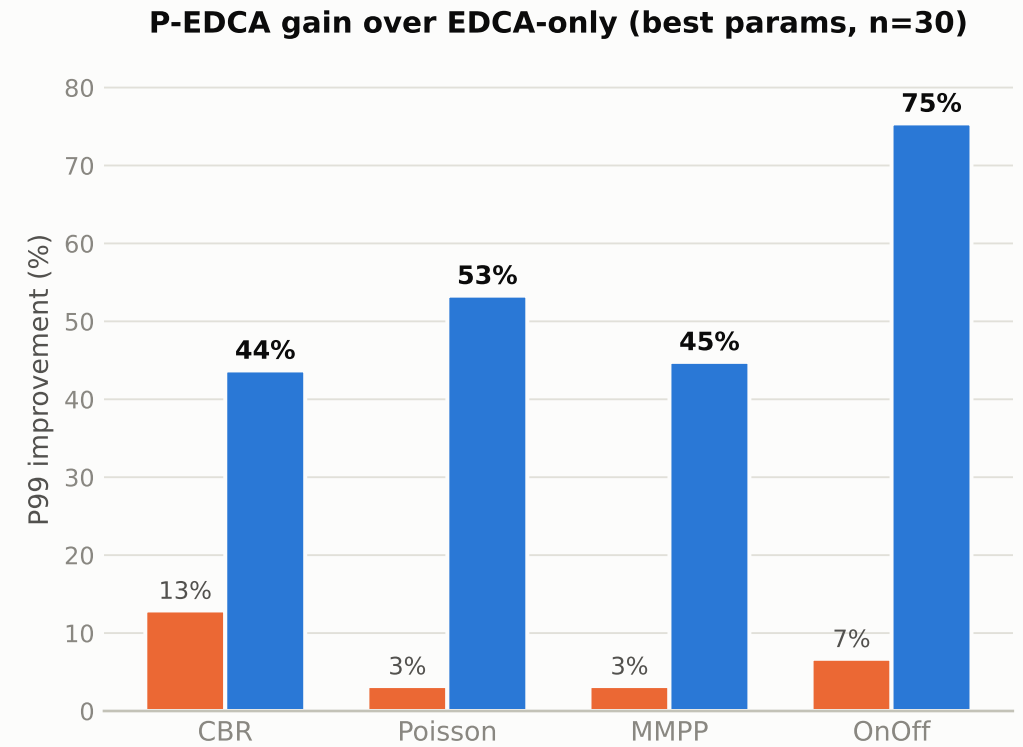
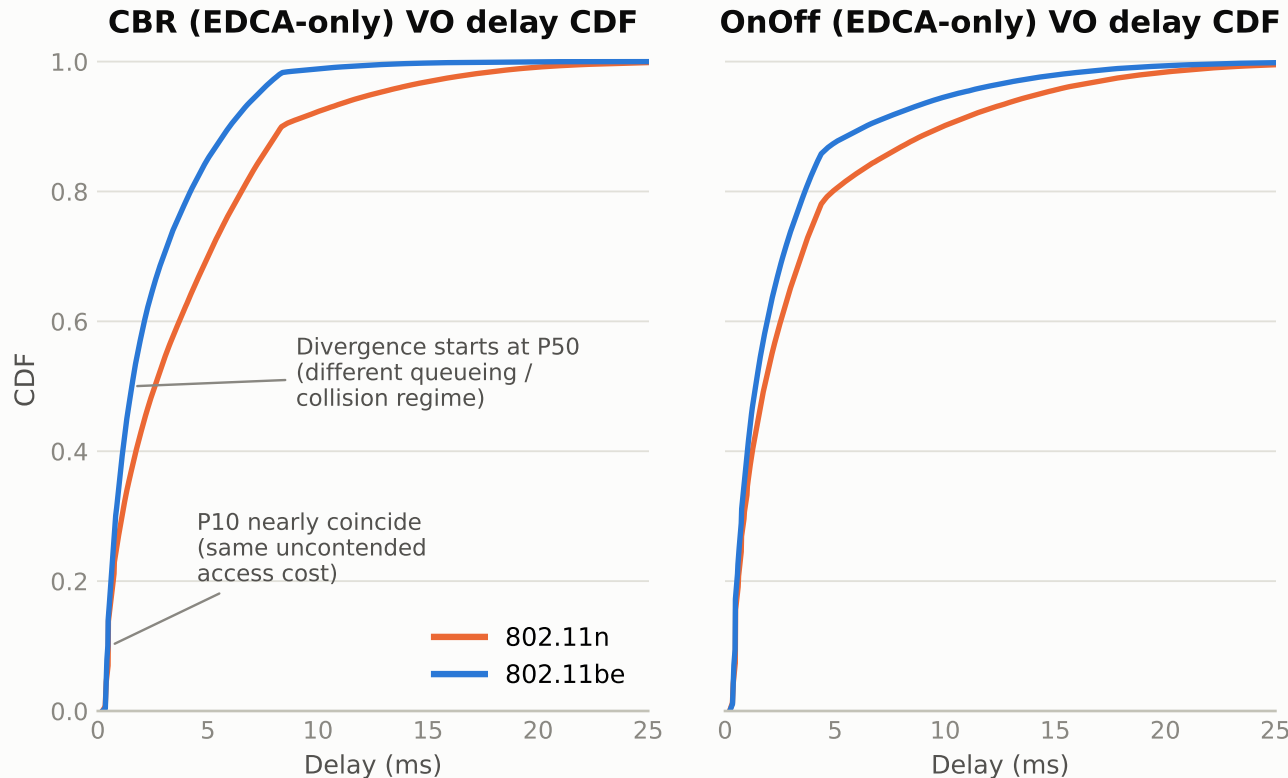
Minimum P99 over the 36-combo sweep, per traffic × nPedca (mean of 10 seeds; ms)



- Light load n=5: near tie (11be -2% to -9%) - the system is uncongested, so the PHY generation barely matters.
- Medium n=15: 11be -28% to -48%. Heavy n=30: 11be -56% to -78% (11n at 17-21 ms, 11be held at 4.5-9.5 ms).
- The one reversal (OnOff n=5, +57%) is a min-statistic artifact: combo medians tie (16.5 vs 16.3 ms), 11be wins 18/36 head-to-head, and 11be wins min-P50 (-18%) and min-P95 (-6%) - only the min-P99 cell flips. Details in the speaker notes / analysis.
- Best combos: 11be prefers mild settings (c0/c1 - q0/q1 - s3); 11n's conservative large-CW picks under load cannot recover -> saturated.

Mechanism: queueing & collisions, not frame speed

The whole delay distribution shifts left. EDCA-only baseline (no P-EDCA, 30 STAs) isolates the two PHY generations.

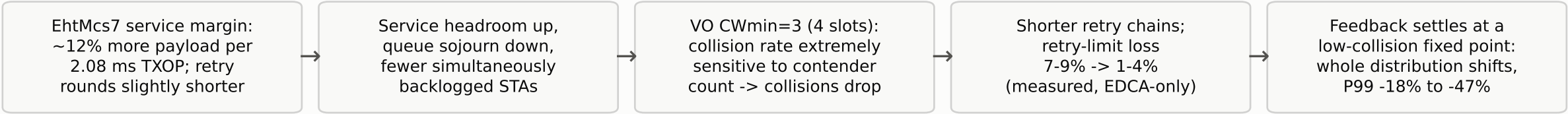


- P10 differs -2% to -4%, P50 -23% to -41%, P99 -18% to -47%: same starting point, gap grows toward the tail.
- PHY gain shrinks with burstiness: EDCA-only P99 gain CBR -47% -> Poisson -30% -> MMPP -22% -> OnOff -18% (burst-scale queueing is arrival-driven).
- P-EDCA view: on loaded 11n the DS-CTS overhead no longer pays (+3-13%); on 11be the same mechanism still has headroom and yields +44-75%.

Root cause: different fixed points of the collision-queueing feedback loop

Per-frame airtime is nearly identical (EHT +12% payload rate, longer preamble) - the cause is the operating point, not raw speed

Established causal chain (11n stuck at the high-collision fixed point; 11be at the low one)



Supporting evidence (EDCA-only, 30 STAs; 11n -> 11be)

Traffic	VO collision loss	Queue delay (µs)	Access delay (µs)	Channel idle
CBR	9.2% → 1.2%	3,636 → 2,223	312 → 244	25% → 27%
Poisson	9.1% → 2.4%	3,904 → 2,346	366 → 294	23% → 25%
MMPP	8.9% → 3.3%	3,904 → 2,623	393 → 324	23% → 25%
OnOff	7.0% → 3.7%	3,174 → 2,333	421 → 347	22% → 25%

Reading: queue delay is ~90% of MAC delay and differs 35-40%; access delay differs only ~20%; idle differs ~2 pp; carried throughput equal

Why each pattern appears

■ Small gain at light load

Feedback region not entered; delay set by a single access (identical for both PHYs).

■ Smaller gain when bursty

Burst peaks far exceed the service rate; queueing is arrival-driven, the margin only shortens the drain time.

■ Largest gap at n=30

DS-CTS adds one 6 Mbps exchange per access; 11n adds it past the knee (blow-up); 11be absorbs it.

Open items (suggested follow-ups)

Whether the +12% service margin alone explains the fixed-point gap, or ns-3's HT vs EHT frame-exchange / receive paths (post-collision EIFS behavior; this repo's v6-v6.3 changes) also contribute: (1) count PhyRxDrop / collision events per standard; (2) rate-sensitivity run with 11be GI = 0.8 µs (86 Mbps) to separate rate effects from code-path effects.